



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

CNS

LAB 4.1 FOOTPRINTING

The process of collecting information about an organization from publicly accessible sources.

Network Reconnaissance using Linux Command Line

1. At the Linux command prompt type the command-line WHOIS query in the following manner and press Enter:
whois www.slashdot.org
2. Record the registrar for your domain.
3. Record the primary and secondary name servers for this domain.
4. Record the Administrative Contact name, address, and phone number for this domain, if displayed.
5. Record the Technical Contact name, address, and phone number for this domain, if displayed.
6. Record the Billing Contact name, address, and phone number for this domain, if displayed.

Inverse Mapping

You can use a web browser to gather inverse mapping information. First, obtain the IP address of the target in order to find the other IP addresses associated with that target. Do this as shown below.

1. At the Linux command prompt type the host command as shown below
host www.slashdot.org

DNS Interrogation with Linux Commands

1. At the Linux command prompt, type **host -t cname www.slashdot.org** and press Enter. The system responds with the corresponding IP address and any aliases if the cname option is used.
2. Record the IP addresses and any aliases corresponding to the entry:
3. You can also reverse the process and look up a domain name from a known address.
4. Linux has a utility that supplies detailed DNS information on addresses. The command **dig** is used much like host. Run the assigned address through dig and note the name servers.
dig www.slashdot.org
5. Another interesting use of the host utility is to examine the mail servers responsible for a particular address or domain name. In order to specify the type of query you are generating, use the -t modifier in host. You first set the type modifier to mx (mail exchange), and then enter the domain name. The system responds with the first three mail exchange servers, unless the system has been configured not to respond to this query.
host -t mx www.strathmore.edu
6. Record the mail servers corresponding to the DNS addresses you entered.

Using ICMP In Linux

1. At the Linux command line, examine the options available by typing **man ping**. Examine the options available.
2. **ping** www.strathmore.edu and then press Enter.
3. The computer continues to generate ICMP echo requests until halted by pressing Ctrl + C.
4. Record the minimum, maximum, and average return times for your ping:
5. The next step is to ping an unreachable host. Type the following: **ping 192.168.1.5**. Press Ctrl +C to stop the ping process.

Using Traceroute

1. At the command prompt type **traceroute** and press Enter to examine the options available.
2. Type traceroute www.slashdot.org and press Enter. Record what you find.